Machine Markets: Current Numerical Evidence Supplement

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This supplement documents the current reconstructed questions and numerical references. Its intervals are exploratory references under the stated assumptions, not certificates of serial confidence coverage, causal learning attribution, source rights for every transformation, market calibration or independent specialist review. Historical literals are preserved in earlier artifacts and superseded by the named reconstructions.

1 Sources, Representation and Protocols

The original 768 Binance study units are reproduced from current official archives with official SHA256 and ZIP CRC checks. Historical download-time hashes were not preserved. All three recorded one-second files match version 1 of Nielsen's primary release byte for byte (uploader-declared CC0). The January CSV now also matches the original siavashraz version 1 CSV byte for byte (uploader-declared MIT), whose provider describes Binance BTCUSDT.P perpetual data and nominal 250-millisecond sampling. The separately matched Parquet distribution declares Apache 2.0. Provider-declared venue/contract is distinct from independent exchange authentication. The FI-2010 primary record declares CC BY 4.0, while independent normalized-file transformation replay matches the supplied CSV, while original archive hash and company/unit authentication remain open. Current Binance Vision terms specify CC BY-NC-SA 4.0; final derivative publication must reflect the provider terms.

The recorder baseline admits occupied ten-second bins, including partial bins. Its EMA resets at missing whole bins and continues across contiguous dates; regressions split dates and bin gaps. The strict sensitivity requires ten observed seconds per bin and resets across the resulting gaps. Recorded flow is a constructed feature; source-byte authentication does not authenticate an executed-volume accumulation window. The original Binance reconstruction separately requires complete bins and resets by date/gap. Equity uses event-indexed supplied labels, not authenticated company identities. January uses top-quote OFI, not executed trades.

2 Strict Recorder Completeness Sensitivity

Strict constructions use EMA 300, burn 150 ten-second states and sufficiently long date/gap segments. Counts and date-Student references are calculated from separately saved segment estimates. Baseline partial-bin counts are 75(BTC), 72(ETH), 69(ADA); removing those bins also changes subsequent EMA burn and segment eligibility.

Table 1: Every strict recorder construction; all questions retained.

Asset	Construction	Dates	Segments	Transitions	Q	Date reference
BTC	x_observed	13	70	92591	0.0010800	[0.0004854, 0.0016746]
BTC	x_forward	13	70	92521	0.0091772	[0.0060301, 0.0123244]
BTC	return_observed	13	70	92591	0.0005381	[-0.0007296, 0.0018059]
ETH	x_observed	13	72	92141	0.0007072	[0.0002962, 0.0011181]
ETH	x_forward	13	72	92069	0.0098205	[0.0075205, 0.0121205]
ETH	return_observed	13	72	92141	0.0003063	[-0.0004946, 0.0011071]
ADA	x_observed	13	68	92874	0.0004148	[-0.0000503, 0.0008798]
ADA	x_forward	13	68	92806	0.0061002	[0.0046502, 0.0075502]
ADA	return_observed	13	68	92874	-0.0001126	[-0.0009599, 0.0007347]

3 Original-Panel Lag Controls and Date References

All 768 study units are reconstructed without filling. L counts ten-second bars, so $L = 90$ is 900 seconds. Segment estimates combine within each date using transition weights. The following intervals use equal-weight sampled dates, with serial-dependence and selection limitations.

Table 2: Full six-horizon original-panel lag ledger.

Pair	L	Seconds	Dates	Mean Q_L	Date reference
BTC	1	10	96	0.0006350	[0.0002804, 0.0009896]
BTC	5	50	96	0.0006233	[0.0002706, 0.0009760]
BTC	15	150	96	0.0005719	[0.0002440, 0.0008997]
BTC	30	300	96	0.0005751	[0.0002396, 0.0009106]
BTC	60	600	96	0.0006352	[0.0002804, 0.0009899]
BTC	90	900	96	0.0006545	[0.0002863, 0.0010227]
ETH	1	10	96	0.0005657	[0.0002710, 0.0008605]
ETH	5	50	96	0.0003970	[0.0001416, 0.0006524]
ETH	15	150	96	0.0003256	[0.0000649, 0.0005864]
ETH	30	300	96	0.0003358	[0.0000699, 0.0006018]
ETH	60	600	96	0.0003685	[0.0000854, 0.0006516]
ETH	90	900	96	0.0003876	[0.0000900, 0.0006852]
SOL	1	10	96	0.0002036	[0.0000323, 0.0003749]
SOL	5	50	96	0.0001401	[0.0000037, 0.0002765]
SOL	15	150	96	0.0001265	[0.0000005, 0.0002524]
SOL	30	300	96	0.0001314	[0.0000015, 0.0002614]
SOL	60	600	96	0.0001270	[-0.0000125, 0.0002664]
SOL	90	900	96	0.0001371	[-0.0000106, 0.0002849]
ADA	1	10	96	0.0003953	[0.0002145, 0.0005761]
ADA	5	50	96	0.0003466	[0.0001729, 0.0005204]
ADA	15	150	96	0.0003536	[0.0001750, 0.0005322]
ADA	30	300	96	0.0003623	[0.0001805, 0.0005440]
ADA	60	600	96	0.0003734	[0.0001815, 0.0005654]
ADA	90	900	96	0.0004001	[0.0001937, 0.0006064]
DOGE	1	10	96	0.0002511	[-0.0000405, 0.0005426]
DOGE	5	50	96	0.0001911	[-0.0001093, 0.0004914]
DOGE	15	150	96	0.0001588	[-0.0001380, 0.0004556]
DOGE	30	300	96	0.0001622	[-0.0001389, 0.0004634]
DOGE	60	600	96	0.0001604	[-0.0001764, 0.0004972]
DOGE	90	900	96	0.0002019	[-0.0001479, 0.0005518]
BNB	1	10	96	0.0004587	[0.0002673, 0.0006501]
BNB	5	50	96	0.0003060	[0.0001251, 0.0004870]
BNB	15	150	96	0.0002888	[0.0001017, 0.0004759]
BNB	30	300	96	0.0002731	[0.0000738, 0.0004724]
BNB	60	600	96	0.0003099	[0.0001071, 0.0005128]
BNB	90	900	96	0.0003253	[0.0001145, 0.0005360]
XRP	1	10	96	0.0002292	[0.0000591, 0.0003994]
XRP	5	50	96	0.0002245	[0.0000642, 0.0003847]
XRP	15	150	96	0.0001978	[0.0000338, 0.0003618]
XRP	30	300	96	0.0001741	[-0.0000057, 0.0003538]
XRP	60	600	96	0.0002046	[0.0000074, 0.0004017]
XRP	90	900	96	0.0002194	[0.0000091, 0.0004297]
LTC	1	10	96	0.0003068	[0.0000419, 0.0005717]
LTC	5	50	96	0.0002120	[0.0000051, 0.0004190]
LTC	15	150	96	0.0002666	[0.0000010, 0.0005322]
LTC	30	300	96	0.0002650	[-0.0000018, 0.0005318]
LTC	60	600	96	0.0002881	[-0.0000043, 0.0005805]
LTC	90	900	96	0.0003046	[0.0000024, 0.0006068]

4 Calendar Dependence Sensitivity

Calendar Bartlett weights depend on actual elapsed days, rather than treating irregular sampled dates as equally spaced. The Gaussian references require additional mixing/stationarity conditions. The 32-reference grid is retrospective and does not establish finite-sample or causal confidence coverage.

Table 3: All calendar covariance references.

Construction	Index	Days	Dates	Mean	Gaussian reference
legacy	Q	30	96	0.0081536	[0.0072655, 0.0090417]
legacy	Q	90	96	0.0081536	[0.0072138, 0.0090934]
legacy	Q	180	96	0.0081536	[0.0072308, 0.0090765]
legacy	Q	365	96	0.0081536	[0.0071973, 0.0091099]
legacy	S	30	96	0.9205544	[0.9157152, 0.9253936]
legacy	S	90	96	0.9205544	[0.9150977, 0.9260111]
legacy	S	180	96	0.9205544	[0.9150249, 0.9260839]
legacy	S	365	96	0.9205544	[0.9145238, 0.9265850]
legacy	Q30	30	96	0.0064012	[0.0054535, 0.0073490]
legacy	Q30	90	96	0.0064012	[0.0053565, 0.0074460]
legacy	Q30	180	96	0.0064012	[0.0053001, 0.0075024]
legacy	Q30	365	96	0.0064012	[0.0051774, 0.0076250]
observed	Q	30	96	0.0003807	[0.0002410, 0.0005203]
observed	Q	90	96	0.0003807	[0.0002204, 0.0005410]
observed	Q	180	96	0.0003807	[0.0002270, 0.0005343]
observed	Q	365	96	0.0003807	[0.0002680, 0.0004934]
observed	S	30	96	0.9203825	[0.9152239, 0.9255411]
observed	S	90	96	0.9203825	[0.9144143, 0.9263508]
observed	S	180	96	0.9203825	[0.9141305, 0.9266346]
observed	S	365	96	0.9203825	[0.9134590, 0.9273060]
observed	Q30	30	96	0.0002849	[0.0001610, 0.0004087]
observed	Q30	90	96	0.0002849	[0.0001512, 0.0004186]
observed	Q30	180	96	0.0002849	[0.0001611, 0.0004087]
observed	Q30	365	96	0.0002849	[0.0001913, 0.0003785]
raw_return	Q	30	96	-0.0001625	[-0.0003858, 0.0000608]
raw_return	Q	90	96	-0.0001625	[-0.0004270, 0.0001020]
raw_return	Q	180	96	-0.0001625	[-0.0004651, 0.0001400]
raw_return	Q	365	96	-0.0001625	[-0.0005076, 0.0001825]
raw_return	S	30	96	-0.0414724	[-0.0531775, -0.0297674]
raw_return	S	90	96	-0.0414724	[-0.0562110, -0.0267339]
raw_return	S	180	96	-0.0414724	[-0.0591562, -0.0237887]
raw_return	S	365	96	-0.0414724	[-0.0623865, -0.0205584]

5 Fee Comparison: Reference Sensitivity

The four-regressor BTC-by-post specification has 192 rows, one treated asset, eight asset groups and 24 date groups. Non-rejection does not establish causal invariance. The legacy placebo rank is not a randomized-treatment probability. The old 0.007 equivalence margin is inappropriate relative to the corrected baseline magnitude.

Table 4: Corrected observed-flow coefficient under all declared references.

Index	Reference	df	Effect	SE	p	Reference interval
Q	classical	188	-0.0001057	0.0004038	0.793794	[-0.0009023, 0.0006909]
Q	asset_cluster	7	-0.0001057	0.0002139	0.636344	[-0.0006116, 0.0004001]
Q	date_cluster	23	-0.0001057	0.0004449	0.814284	[-0.0010260, 0.0008146]
S	classical	188	-0.0036464	0.0254227	0.886102	[-0.0537968, 0.0465039]
S	asset_cluster	7	-0.0036464	0.0203732	0.863022	[-0.0518215, 0.0445286]
S	date_cluster	23	-0.0036464	0.0140420	0.797420	[-0.0326946, 0.0254017]

6 Covariance Model Restrictions

All fixed centering/bandwidth cases are retained, with six starts each. The model uses unrestricted stable transition and positive-definite innovation parameterization. The 11 moments and seven parameters give a four-degree asymptotic reference. Rejection is evidence against maintained restrictions for that construction, not validation of latent learning or a causal model.

Table 5: Complete maintained-model diagnostic grid.

Asset	Case	Rows	df	J	p	Q
BTC	global_5	102454	4	44.15380	5.961e-09	0.0009757
BTC	global_10	102454	4	41.40631	2.214e-08	0.0008314
BTC	global_20	102454	4	42.12612	1.571e-08	0.0005814
BTC	global_40	102454	4	44.11380	6.076e-09	0.0003780
BTC	within_segment_5	102454	4	43.36879	8.676e-09	0.0009871
BTC	within_segment_10	102454	4	40.60477	3.245e-08	0.0008423
BTC	within_segment_20	102454	4	41.30951	2.319e-08	0.0005942
BTC	within_segment_40	102454	4	43.31609	8.897e-09	0.0003916
ETH	global_5	102757	4	29.01753	7.754e-06	0.0006356
ETH	global_10	102757	4	28.30939	1.08e-05	0.0004755
ETH	global_20	102757	4	27.12692	1.874e-05	0.0003755
ETH	global_40	102757	4	25.39847	4.183e-05	0.0003232
ETH	within_segment_5	102757	4	28.78851	8.63e-06	0.0006313
ETH	within_segment_10	102757	4	28.11647	1.181e-05	0.0004721
ETH	within_segment_20	102757	4	26.98036	2.006e-05	0.0003725
ETH	within_segment_40	102757	4	25.28746	4.404e-05	0.0003212
ADA	global_5	102885	4	13.12143	0.0107	0.0009049
ADA	global_10	102885	4	12.11749	0.0165	0.0009106
ADA	global_20	102885	4	11.47824	0.02168	0.0008940
ADA	global_40	102885	4	11.30958	0.0233	0.0008720
ADA	within_segment_5	102885	4	12.62822	0.01324	0.0008992
ADA	within_segment_10	102885	4	11.67935	0.0199	0.0009050
ADA	within_segment_20	102885	4	11.07393	0.02575	0.0008885
ADA	within_segment_40	102885	4	10.90896	0.02761	0.0008662

7 Joint Cross-Venue Association

The joint four-variable VAR uses 11 matched April 2021 dates, complete ten-second bins, segment effects and actual timestamps. Generalized shares are normalized by row. All 24 orthogonal orderings are retained. Quote-currency and timing differences and missing structural shock identification prevent a causal transmission claim.

Table 6: All six horizons for both predictive directions.

Asset	Seconds	ρ	BN-to-recorder	Ordering range	Recorder-to-BN
BTC	10	0.9506137	0.31381	[0.00000, 0.54690]	0.31919
BTC	30	0.9506137	0.38934	[0.07367, 0.74437]	0.32098
BTC	60	0.9506137	0.44293	[0.16997, 0.84831]	0.32244
BTC	100	0.9506137	0.47145	[0.22810, 0.89275]	0.32330
BTC	300	0.9506137	0.49634	[0.27931, 0.92808]	0.32410
BTC	600	0.9506137	0.49859	[0.28388, 0.93120]	0.32417
ETH	10	0.9499339	0.35660	[0.00000, 0.62957]	0.35535
ETH	30	0.9499339	0.38599	[0.01592, 0.72413]	0.36456
ETH	60	0.9499339	0.41315	[0.04774, 0.79463]	0.37446
ETH	100	0.9499339	0.43162	[0.07663, 0.83657]	0.38144
ETH	300	0.9499339	0.45121	[0.11116, 0.87755]	0.38900
ETH	600	0.9499339	0.45306	[0.11450, 0.88135]	0.38973
ADA	10	0.9482043	0.30252	[0.00000, 0.52603]	0.29781
ADA	30	0.9482043	0.31190	[0.00156, 0.56829]	0.30098

Asset	Seconds	ρ	BN-to-recorder	Ordering range	Recorder-to-BN
ADA	60	0.9482043	0.32520	[0.00701, 0.60873]	0.30750
ADA	100	0.9482043	0.33859	[0.01625, 0.64525]	0.31430
ADA	300	0.9482043	0.36344	[0.04370, 0.70612]	0.32692
ADA	600	0.9482043	0.36706	[0.04893, 0.71414]	0.32876

8 Switching Models and Forecast Boundaries

The scalar switching regression uses exact joint transition posteriors, four starts, two variance floors, two specifications and six datasets. Forecasts use seven chronological expanding test dates, April 13–19, 2021. All MSE references span zero; every full-coefficient versus variance-only reference also spans zero. Selected iteration-limit fits and variance-floor sensitivity remain part of the result.

Table 7: Every reported switching forecast comparison.

Dataset	Comparison	MSE change	Date reference	Density change	Date reference
BTC_x	Var–OLS	-0.10456	[-0.31673, 0.10760]	0.69176	[0.57185, 0.81167]
BTC_x	Full–OLS	-0.00411	[-0.01376, 0.00554]	0.69655	[0.56985, 0.82325]
BTC_x	Full–Var	0.10045	[-0.10222, 0.30312]	0.00480	[-0.00493, 0.01452]
BTC_return	Var–OLS	-0.19435	[-0.53914, 0.15044]	0.69008	[0.57227, 0.80790]
BTC_return	Full–OLS	-0.00774	[-0.02285, 0.00737]	0.69750	[0.57129, 0.82370]
BTC_return	Full–Var	0.18661	[-0.14317, 0.51639]	0.00741	[-0.00582, 0.02064]
ETH_x	Var–OLS	0.00344	[-0.02834, 0.03523]	0.22412	[0.08228, 0.36595]
ETH_x	Full–OLS	0.00133	[-0.00110, 0.00375]	0.22432	[0.08143, 0.36720]
ETH_x	Full–Var	-0.00212	[-0.03403, 0.02979]	0.00020	[-0.00112, 0.00152]
ETH_return	Var–OLS	-0.02036	[-0.07868, 0.03795]	0.22284	[0.08443, 0.36126]
ETH_return	Full–OLS	0.00035	[-0.00442, 0.00512]	0.22258	[0.08405, 0.36112]
ETH_return	Full–Var	0.02072	[-0.03542, 0.07685]	-0.00026	[-0.00148, 0.00096]
ADA_x	Var–OLS	-0.03059	[-0.29351, 0.23233]	0.40333	[-0.06704, 0.87371]
ADA_x	Full–OLS	-0.03785	[-0.10228, 0.02659]	0.40278	[-0.06553, 0.87109]
ADA_x	Full–Var	-0.00726	[-0.33236, 0.31785]	-0.00055	[-0.00439, 0.00328]
ADA_return	Var–OLS	-0.05595	[-0.15889, 0.04699]	0.40255	[-0.06614, 0.87125]
ADA_return	Full–OLS	0.00726	[-0.02590, 0.04043]	0.40273	[-0.06633, 0.87178]
ADA_return	Full–Var	0.06322	[-0.06803, 0.19447]	0.00017	[-0.00089, 0.00123]

9 Replayed Financial Scenarios

The 22,000 existing funded paths are rerun with their saved seed and independently reconstructed cash/mark identity. Capacity scales the 25-share inventory limit at fixed initial equity 10,000. Effective anchor is 0.80 rather than nominal 0.40. All effective systems in that grid are supercritical; inventory-limit hits are sticky occurrences, not a permanent ban or default. Independent inventory projection sensitivity leaves the reported hit counts unchanged.

Table 8: All 22 funded cells, 1,000 paths each.

Nominal S	Effective S	Capacity	Shock	Hits	Defaults	Mean minimum equity
0.70	1.10	0.5	-0.2	691	0	9998.9458
0.70	1.10	0.5	-0.6	1000	0	9999.9521
0.70	1.10	1.0	-0.2	0	0	9999.1197
0.70	1.10	1.0	-0.6	0	0	9998.6558
0.70	1.10	2.0	-0.6	0	0	9997.4167
0.70	1.10	2.0	-2.0	0	0	9983.7661
0.82	1.22	0.5	-0.2	1000	0	9999.5624
0.82	1.22	0.5	-0.4	1000	0	9999.5605
0.82	1.22	1.0	-0.2	1000	0	9995.3513
0.82	1.22	1.0	-0.4	1000	0	9996.4432
0.82	1.22	1.0	-0.6	1000	0	9998.4149

Nominal S	Effective S	Capacity	Shock	Hits	Defaults	Mean minimum equity
0.82	1.22	1.0	-0.8	1000	0	9999.9079
0.82	1.22	2.0	-0.6	1000	0	9978.2319
0.82	1.22	2.0	-1.0	1000	0	9979.7946
0.82	1.22	2.0	-2.0	1000	0	9985.6240
1.18	1.58	0.5	-0.2	1000	0	9998.7996
1.18	1.58	0.5	-0.4	1000	0	9998.8000
1.18	1.58	1.0	-0.2	1000	0	9994.9766
1.18	1.58	1.0	-0.6	1000	0	9994.9768
1.18	1.58	1.0	-0.8	1000	0	9994.9768
1.18	1.58	2.0	-1.0	1000	0	9979.7590
1.18	1.58	2.0	-2.0	1000	0	9979.7559

The separate lagged-memory dealer experiment uses 400 paths per cell. Its archived halt flag combines nonpositive equity and a position-limit event. These causes are not separated in the saved count, and its flow calibration remains model moments rather than real queue fills. All 48 cells and exact binomial references are retained below.

Table 9: Complete conditional dealer replay; a limit event is not an observed market default.

S	Leverage	Mode	Halts	Paths	Rate	Exact binomial reference
0.50	10	feedback	44	400	0.1100	[0.08107, 0.14485]
0.50	10	B1	45	400	0.1125	[0.08325, 0.14763]
0.50	10	B2	42	400	0.1050	[0.07673, 0.13927]
0.50	10	B3	43	400	0.1075	[0.07890, 0.14206]
0.50	20	feedback	0	400	0.0000	[0.00000, 0.00918]
0.50	20	B1	0	400	0.0000	[0.00000, 0.00918]
0.50	20	B2	0	400	0.0000	[0.00000, 0.00918]
0.50	20	B3	0	400	0.0000	[0.00000, 0.00918]
0.60	10	feedback	118	400	0.2950	[0.25072, 0.34234]
0.60	10	B1	46	400	0.1150	[0.08544, 0.15040]
0.60	10	B2	45	400	0.1125	[0.08325, 0.14763]
0.60	10	B3	126	400	0.3150	[0.26974, 0.36302]
0.60	20	feedback	0	400	0.0000	[0.00000, 0.00918]
0.60	20	B1	0	400	0.0000	[0.00000, 0.00918]
0.60	20	B2	0	400	0.0000	[0.00000, 0.00918]
0.60	20	B3	0	400	0.0000	[0.00000, 0.00918]
0.70	10	feedback	249	400	0.6225	[0.57298, 0.67019]
0.70	10	B1	52	400	0.1300	[0.09864, 0.16696]
0.70	10	B2	56	400	0.1400	[0.10752, 0.17792]
0.70	10	B3	276	400	0.6900	[0.64214, 0.73502]
0.70	20	feedback	1	400	0.0025	[0.00006, 0.01385]
0.70	20	B1	0	400	0.0000	[0.00000, 0.00918]
0.70	20	B2	0	400	0.0000	[0.00000, 0.00918]
0.70	20	B3	1	400	0.0025	[0.00006, 0.01385]
0.80	10	feedback	373	400	0.9325	[0.90331, 0.95505]
0.80	10	B1	59	400	0.1475	[0.11422, 0.18610]
0.80	10	B2	89	400	0.2225	[0.18267, 0.26649]
0.80	10	B3	391	400	0.9775	[0.95772, 0.98966]
0.80	20	feedback	22	400	0.0550	[0.03479, 0.08209]
0.80	20	B1	0	400	0.0000	[0.00000, 0.00918]
0.80	20	B2	0	400	0.0000	[0.00000, 0.00918]
0.80	20	B3	40	400	0.1000	[0.07241, 0.13368]
0.90	10	feedback	400	400	1.0000	[0.99082, 1.00000]
0.90	10	B1	106	400	0.2650	[0.22238, 0.31114]
0.90	10	B2	188	400	0.4700	[0.42022, 0.52023]
0.90	10	B3	400	400	1.0000	[0.99082, 1.00000]
0.90	20	feedback	210	400	0.5250	[0.47478, 0.57484]
0.90	20	B1	0	400	0.0000	[0.00000, 0.00918]

S	Leverage	Mode	Halts	Paths	Rate	Exact binomial reference
0.90	20	B2	0	400	0.0000	[0.00000, 0.00918]
0.90	20	B3	258	400	0.6450	[0.59592, 0.69192]
0.95	10	feedback	400	400	1.0000	[0.99082, 1.00000]
0.95	10	B1	211	400	0.5275	[0.47728, 0.57731]
0.95	10	B2	350	400	0.8750	[0.83854, 0.90578]
0.95	10	B3	400	400	1.0000	[0.99082, 1.00000]
0.95	20	feedback	353	400	0.8825	[0.84683, 0.91238]
0.95	20	B1	0	400	0.0000	[0.00000, 0.00918]
0.95	20	B2	12	400	0.0300	[0.01560, 0.05182]
0.95	20	B3	385	400	0.9625	[0.93890, 0.97886]

10 January Retrospective Forecast Reconstruction

The existing authenticated January sample now has its own forecast computation. Quote OFI and the gap-reset EMA 120 retain the same 152 eligible date/gap segments and 844,621 baseline transitions. Regression lags split dates/gaps. Six initial training dates precede each of six expanding evaluation dates, January 15–20, 2023. All four lag horizons are retained; normalization and coefficients use training observations only. The one-second lag grid and protocol are retrospective, rather than an untouched holdout or preregistered selection. Positive mean reductions are predictive associations, with six serial dates and no multiple-comparison or causal guarantee.

Table 10: All existing-sample January AR-minus-VAR losses; scale is training SD.

Lag sec	Target	Dates	AR MSE	Mean reduction	Date reference	Reduction percent
1	price	6	0.0203504	0.0003750	[0.0003470, 0.0004030]	1.84275
1	flow	6	0.7218236	0.0001187	[-0.0000583, 0.0002958]	0.01645
2	price	6	0.0203144	0.0003624	[0.0002907, 0.0004341]	1.78394
2	flow	6	0.7195081	0.0008158	[0.0005398, 0.0010918]	0.11339
3	price	6	0.0203085	0.0003608	[0.0002840, 0.0004377]	1.77683
3	flow	6	0.7185951	0.0007943	[0.0005271, 0.0010614]	0.11053
6	price	6	0.0203145	0.0003574	[0.0002781, 0.0004367]	1.75939
6	flow	6	0.7179558	0.0007489	[0.0004968, 0.0010011]	0.10432

11 Equity Source Replay and Full Coverage

The supplied CSV now reproduces byte for byte from ten independently retrieved normalized DecPre files using its preserved transformation. The six-largest-jump/five-interval rule omits the endpoint: it retains 272,941 of 394,337 events and drops 121,396 (30.7848 percent). Legacy labels 2–5 map to different inferred price blocks across days and are not authenticated company identities. A full-coverage sensitivity uses four largest finite price gaps plus an explicit endpoint, retaining all 394,337 events in 50 eligible day/block sessions. The event EMA 100, burn 500 and preserved decimal conversion remain explicit. All 50 fits are independently reconstructed using decimal GCD, manual OFI/EMA and QR. The smallest selected gap is at least 12.3055 times the largest omitted gap; this supports a declared partition, not a company map. Primary Fairdata archive byte identity and original physical normalization denominators remain unverified.

Table 11: All full-coverage blocks and ten-day references; block 1 Q spans zero.

Block	Index	Events	Sessions	Mean	Date reference
1	Q	29744	10	0.0001936	[-0.0000045, 0.0003918]
1	S	29744	10	0.9783431	[0.9746987, 0.9819875]
2	Q	67935	10	0.0004265	[0.0002795, 0.0005734]
2	S	67935	10	0.9782194	[0.9753796, 0.9810593]
3	Q	62898	10	0.0005447	[0.0003273, 0.0007622]

Block	Index	Events	Sessions	Mean	Date reference
3	S	62898	10	0.9748446	[0.9685557, 0.9811334]
4	Q	86605	10	0.0002904	[0.0001552, 0.0004256]
4	S	86605	10	0.9772303	[0.9751638, 0.9792968]
5	Q	147155	10	0.0001433	[0.0000254, 0.0002613]
5	S	147155	10	0.9824020	[0.9793084, 0.9854956]

12 Financial Information and Publication Boundaries

The main manuscript and central-result review packet now specify the physical input, positive price-response geometry, refreshed average-book execution, actual cash/inventory and distinct counterparty/probe-trader funding. A restricted noisy-price observer faces the stated resource orders. Exact marked equity and naturally excited price blocks change that experiment. These constructed financial models do not authenticate actual impact, queue fills, funding, collateral, positions or a financial data-noise contract for the recorded samples.

Existing learned-policy aggregate outputs are audited without new training: their asserted high R-squared conclusion is contradicted by the saved values and constant-action substitution, and the source lacks a sufficient checkpoint chain and genuine queue-fill calibration. The current manuscript uses none of those outputs as an operational safety certificate.

The historical flash-crash script assigns or simulates its coefficients, contains no historical TAQ/account ingest and changes several counterfactual inputs at once. Its underlying research question remains open, and its original outputs are retained as illustrations. Original equity archive byte authentication, company/unit mapping, real financial calibration, defensible causal inference where needed and independent specialist assessment remain publication gates.

Named source/evidence receipts accompany this supplement; a computation, source hash or clean PDF is not scientific or publication closure.

13 Recovered Saved-Policy Response Records

A separate simulated continuous-trading study supplies 20 saved DQN and actor checkpoints. Safe tensor loading, full checkpoint hashes and archive CRC checks precede independent feedforward evaluation. All 100 radius/seed response records and ten seed-aggregate summaries match; the maximum numerical difference is below 10^{-12}. At the smallest radius, actor response R^2 ranges from 0.6540 to 0.9988. Thus the saved conclusion that all neural responses have $R^2 < 0.70$ is contradicted by its own recorded networks.

The source queries price deviation and return together along one diagonal state slice, with inventory and other features fixed. Its `in_linear_class` label is a static fit/slope heuristic, not a dynamic exponential-memory proof. These checkpoints belong to the separate simulated study, not the earlier real-LOB policy implementation. Training reward/loss histories, optimizer trajectories, real fills and operational validity are not established by this response replay. No new training was performed.